UNIVERSITY OF PLYMOUTH

PUSL3190 Computing Project

Project Proposal

TapSync: NFC Event Access and Safety Management System

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Degree Program: BSc (Hons) Software Engineering

PROJECT PROPOSAL

***Do not exceed 2 pages**

1. PROJECT TITLE	TapSync: NFC Event Access and Safety Management System	
2. DATE	**11/7/2026**	
3. PREPARED BY	**NAME:RMSD Rathnayaka**	**INDEX: 10967199**
4. SUPERVISOR NAME	Ms. Dharani Ranasinghe	

5. PROBLEM STATEMENT

Event organizers often manage registrations, credentials, entry points, staff permissions, and emergency attendance through separate processes. This can cause long queues, duplicate or invalid credentials, slow responses to changing access requirements, and incomplete information during incidents. A centralized system with offline-capable gate checks and a shared operational dashboard can address these problems.

6. PROPOSED SOLUTION

TapSync will consist of a desktop administration hub, a gate application, an NFC credential workflow, a backend synchronization service, and an operations dashboard. Organizers create an event and define attendee categories, access zones, and staff roles. Attendees are registered individually or imported in bulk; credentials are then encoded and bound to attendee records.

At entry points, staff tap a credential using a supported NFC reader. The gate application checks cached credential and access-rule data locally, provides an admit or deny result, and records the decision for synchronization later. The administration hub and dashboard provide event staff with attendance summaries, access activity, zone information, and roll call tools

7. OBJECTIVES & DELIVERABLES

- Provide a centralized desktop hub for event setup, attendee records, credential encoding, and staff administration.
- Support attendee registration, spreadsheet import, check-in, and NFC credential assignment.
- Validate credentials at gates against access tiers and zone permissions, including when offline.
- Detect duplicate, revoked, or suspicious credentials and keep an auditable record of access decisions.
- Give organizers a live attendance and zone dashboard with operational alerts.
- Provide an emergency roll-call workflow using the latest available attendance and zone data.
- Evaluate the prototype through functional, usability, offline-operation, and security testing.

8. TARGET USERS

- Event organizers and administrators
- Registration and box-office staff
- Emergency response coordinators
- Security supervisors
- Gate and ground staff

9. CORE FEATURES

- Centralized Event Administration
- Attendee Registration and Check in
- NFC Credential Management
- Gate Access Validation
- Offline-first Operation and Synchronization
- Access Tiers, Zones, and Entry Rules
- Credential Integrity and Supervisor Overrides
- Live Operations Dashboard and Alerts
- Emergency Roll Call
- Audit Trail and Attendance Reports

10. TECHNOLOGIES TO BE USED

- Desktop Application: Tauri (Rust/React)
- Mobile Apps: React Native or Flutter
- Backend: Node.js, Express, PostgreSQL
- Hardware: ISO/IEC 14443 NFC, PC/SC readers, NDEF
- Web: HTML, CSS, Blazor, Aspire

11. NOVELTY

The project will use an iterative development approach. Requirements will be confirmed with representative event roles, followed by system and data design. A working prototype will be developed in increments, beginning with event setup, registration, credential encoding, and gate validation. Dashboard, synchronization, and emergency workflows will then be integrated. Each increment will be reviewed against defined acceptance criteria and revised using feedback from trial users.

12. TIMELINE & MILESTONES

Phase	Milestone / Task	Estimated Completion
1. Requirements and planning	Confirm cases, requirements, constraints, and acceptance criteria	2 weeks
2. Design	Architecture, data model, NFC credential format, and interface prototypes	3 weeks
3. Core prototype	Registration, staff setup, credential encoding, and gate validation	5 weeks
4. Operations features	Dashboard, offline synchronization, access rules, and roll call	5 weeks
5. Evaluation and delivery	User evaluation, corrections, documentation, and demonstration	3 weeks

Supervisor Approval

Documentations	Proposal	Interim 01	Interim 02	Final Submission
Date	11/06			
Approved (Yes / No)	Yes			
Student Signature				
Supervisor Signature				

Other Comments (Supervisor Use Only)

Student Declaration of AI Tool use in this Assessment

Please indicate your level of usage of generative AI for this assessment - please tick the appropriate category(s).

If the "Assisted Work" or "Partnered Work" category is selected, please expand on the usage and in which elements of the assignment the usage refers to.

Solo Work	**S1 - Generative AI tools have not been used for this assessment.**	☐
Assisted Work	**A1 – Idea Generation and Problem Exploration** Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.	☐
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.	☒
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.	☐
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions. The interpretation and integration of research into the assignment remain the student's responsibility.	☒
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.	☐
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.	☐
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.	☐
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or	☐

	suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.	
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.	☐
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.	☐
	A11 - Other uses not listed above Please specify:	☐
Partnered Work	**P1 - Generative AI tool usage has been used integrally for this assessment** Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:	☐

Please provide details of AI usage and which elements of the coursework this relates to:

I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.	☒
I confirm that all details provide above are an accurate description of how AI was used for this assessment.	☒